Simulating Ocean Alkalinity Enhancement along the European Coast in an Earth System Model

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Background

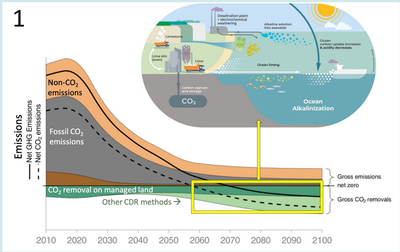
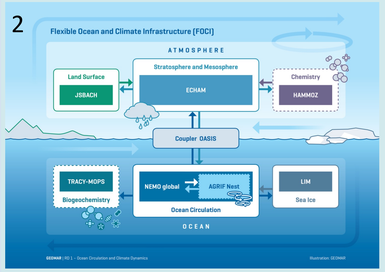


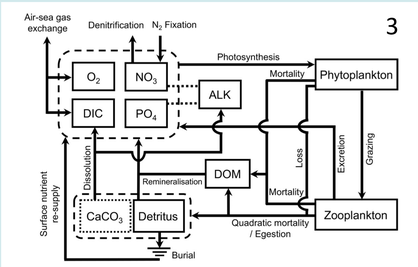
Figure adapted from IPCC AR6 WGIII, Chapter 12, Cross-Chapter Box 8, Figure 2

- In addition to rapid and deep emission reductions, carbon dioxide removal is essential to reach and sustain net zero CO_2 emissions. [1.]
- Sequestering atmospheric carbon dioxide by ocean alkalinity enhancement (OAE) is a proposed method with theorised large scale potential.

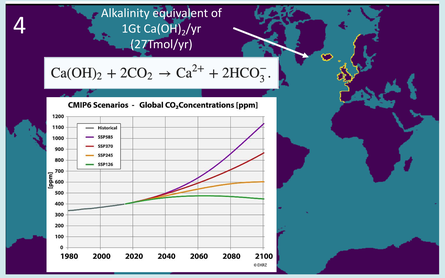
Model Set-up and Experiment



Schematic of earth system model FOCI
Matthes et al. 2020, Figure 1. [2.]



Schematic of the ocean biogeochemistry model in FOCI
Chien et al. 2022, Figure 1. [3.]



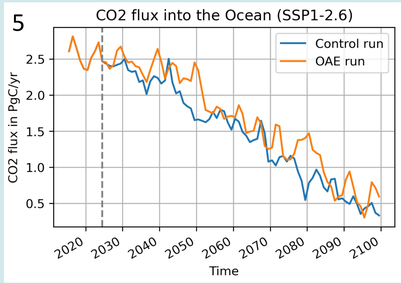
Experiment	Description
Control run	Emissions driven SSP 1-2.6 and SSP 3-7.0 runs from 2015 - 2100
OAE run	Indentical to control with 27 Tmol/yr alkalinity addition along the European coast (50km width) from 2035-2100. Period 2025-2035 has linearly increasing addition.

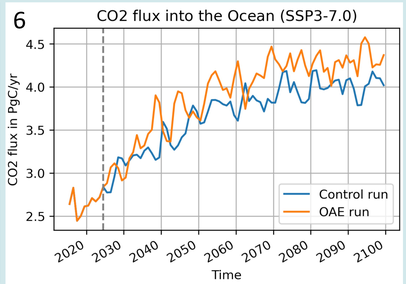
Key questions -

1. How much carbon is taken up by coastal alkalinity enhancement of 1Gt $Ca(OH)_2$/yr?
2. Are there regional differences at the European coast?
3. Does the background climate state affect these results?

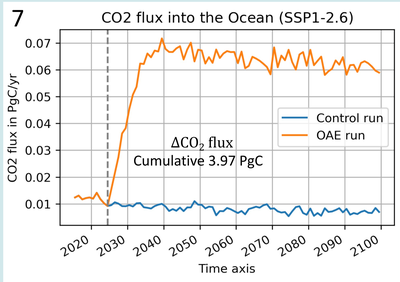
Results - uptake of CO_2

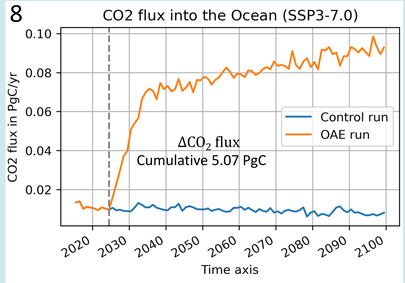
Flux of CO_2 into the ocean - Global



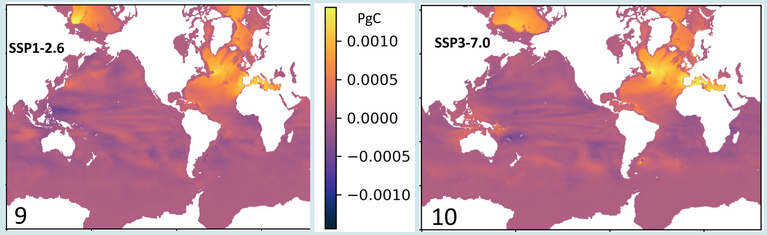


Flux of CO_2 into the ocean - in the region of Alkalinity Enhancement

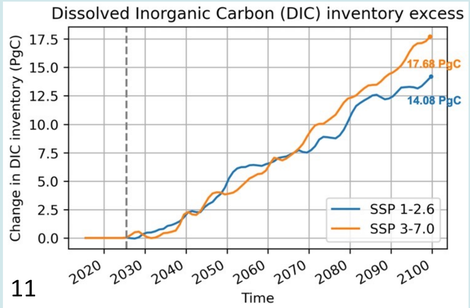




Spatial pattern of change in DIC inventory at the last simulation year



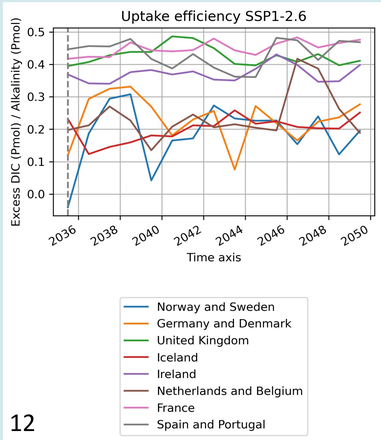
- CO_2 flux into the ocean decreases with time under SSP1-2.6 and increases with time under SSP3-7.0 background climate state.
- In the coastal region of alkalinity enhancement, the cumulative carbon sequestration is 3.97 PgC and 5.07 PgC respectively under SSP1-2.6 and SSP3-7.0.
- Most of the excess in inventory of Dissolved Inorganic Carbon (DIC) stays in the Northern Hemisphere.

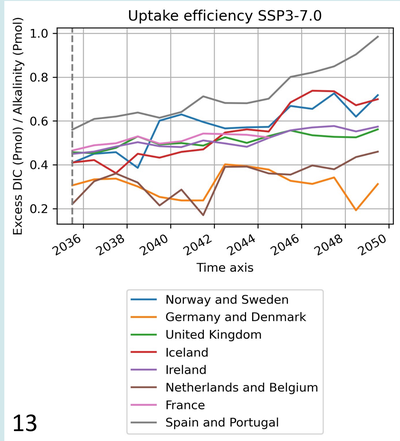


Change in DIC inventory over time (OAE run - Control run)

- The increase in total global DIC shows uptake of CO_2 at different rates in each background climate state.
- For SSP1-2.6, 28.2% of the additional air-sea CO2 flux occurs over the OAE region. This is only slightly larger for SSP3-7.0 at 28.67%.

Results - Regional differences





Key Takeaways-

1. Coastal alkalinity enhancement of 1Gt $Ca(OH)_2$ /yr at the European coast sequestered 17.68 PgC under SSP3-7.0 and 14.08 PgC under SSP1-2.6.
2. There are regional differences in uptake efficiency along the European coast.
3. The background climate state plays a role in the magnitude of carbon uptake but regional partitioning might be similar in both.

References

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